HDOC DAY-58

Question

Write a C program to find the kth smallest element in a variable-length integer array using pointers and bubble sort. Consider duplicate elements separately and display -1 if k is invalid.

Aim

To write a C program to find the kth smallest element in an array using pointers, functions, and bubble sort

Algorithm

1. Start.
2. Read the size n of the array.
3. Declare a variable-length array arr[n].
4. Read n array elements and the value of k.
5. Pass the array to kth_smallest() using a pointer.
6. Check if k is invalid. If yes, print -1.
7. Sort the array in ascending order using bubble sort.
8. Print the element at index k-1.
9. Stop.

>C code

```
#include <stdio.h>

void kth_smallest(int *arr, int n, int k) {
   if (k <= 0 || k > n) {
      printf("-1");
      return;
   }

   int temp;
   for (int i = 0; i < n-1; i++) {
      for (int j = 0; j < n-i-1; j++) {
         if (arr[j] > arr[j+1]) {
            temp = arr[j];
```

```
            arr[j] = arr[j+1];
            arr[j+1] = temp;
        }
    }
}

    printf("kth smallest element: %d", arr[k-1]);
}

int main() {
    int n, k;
    scanf("%d", &n);
    if (n <= 0) {
        printf("-1");
        return 0;
    }

    int arr[n];
    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    scanf("%d", &k);
    kth_smallest(arr, n, k);

    return 0;
}
```